

# Cyrille E. Mvomo

McGill University  
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## Research Interests

My research is on “**interpretables**,” intelligent wearable sensing systems that reveal age- and disease-related deviations in a person’s neurobehavioral state and provide clinicians with information they can use to guide clinical decisions. As a clinician-informatician, I focus on gait and balance dysfunctions. Specifically, my research encompasses **(1) person-specific modeling of neuromotor deficits** to understand the unique presentation of gait and balance impairments in each individual (e.g., via ordinal trends canonical variates analysis); **(2) movement-based biomarker development** to identify signatures inferred from wearables that are associated with the modeled neuromotor deficits (e.g., Lyapunov exponents as proxies for altered gait automaticity); and **(3) clinical decision support system design** to translate these biomarkers into technologies that clinicians can trust to personalize support in routine care and daily life (e.g., reinforcement learning agents).

## Education

|                 |                                                                                                                                                                                                                                                                                                                                                                               |                                             |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| 2027 (Expected) | <b>Ph.D.</b> in Biomechanics and Neurosciences<br>Dissertation: <i>Simple Walking Reveals Less, Complexity Provides Insight: Advancing Digital Gait Biomarkers in Parkinson’s Disease</i><br>Committee: <a href="#">Caroline Paquette</a> (chair), <a href="#">Chris Awai</a> and <a href="#">Phil Dixon</a><br>cGPA: <a href="#">4.0/4.0</a> ( <i>coursework completed</i> ) | McGill University<br>Montreal, Canada       |
| 2023            | <b>Visiting Student</b> in Bioengineering – Robotics<br>Advisors: <a href="#">Mohamed Bouri</a> and <a href="#">Alireza Manzoori</a><br>Project GPA: <a href="#">4.0/4.0</a> [ <i>certificate</i> ]                                                                                                                                                                           | EPFL<br>Lausanne, Switzerland               |
| 2023            | <b>M.Sc.</b> in Informatics (Digital Sciences)<br><i>Concentration in Biomedical Data Science</i><br>cGPA: <a href="#">4.0/4.0</a> [ <i>certificate</i> ]                                                                                                                                                                                                                     | University of Paris – Cité<br>Paris, France |
| 2021            | <b>Bachelor of Podiatric Medicine</b> ( <i>Clinical Degree</i> )<br><b>B.Sc.</b> Grade in Allied Health Sciences ( <i>University Title</i> )<br>cGPA: <a href="#">3.7/4.0</a> [ <i>certificate</i> ]                                                                                                                                                                          | University of Paris – Cité<br>Paris, France |

## Honors and Awards

24 recognitions for excellence in research, teaching, and service, spanning institutional to international levels.

### Fellowships and Scholarships (*all competitive awards independently secured*)

|      |                                                                                                                                                       |                            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2026 | <a href="#">AgeTech Fellowship Top-Up Award, Perley Health</a> (CA\$2,500)                                                                            | <i>Institutional Level</i> |
| 2026 | <a href="#">Graduate Student Award, Parkinson Canada</a> (CA\$40,000)                                                                                 | <i>Federal level</i>       |
| 2026 | <a href="#">Canada Graduate Research Scholarship (CGRS) – Doctoral, Canadian Institutes of Health Research (CIHR)</a> (winning application withdrawn) | <i>Federal level</i>       |
| 2026 | <a href="#">Doctoral Research Scholarship, Quebec Research Fund</a> (CA\$58,333)                                                                      | <i>Provincial level</i>    |
| 2024 | <a href="#">Recruitment Fellowship, Quebec BioImaging Network</a> (CA\$10,000)                                                                        | <i>Provincial level</i>    |
| 2023 | <a href="#">Graduate Excellence Fellowship, McGill University</a> (CA\$60,000)                                                                        | <i>Institutional level</i> |

|      |                                                                                    |                            |
|------|------------------------------------------------------------------------------------|----------------------------|
| 2023 | <a href="#">Anti-Black Racism Excellence Award, McGill University (CA\$10,000)</a> | <i>Non-competitive</i>     |
| 2023 | <a href="#">SMARTS-UP Scholarship, University of Paris – Cité (~CA\$7,800)</a>     | <i>Institutional level</i> |
| 2022 | <a href="#">Visiting Scholarship, Lake Lucerne Institute (~CA\$8,700)</a>          | <i>Non-competitive</i>     |

### Best Papers and Presentations

|      |                                                                                                                                                                                                      |                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2025 | <a href="#">IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI) 2025 – 3<sup>rd</sup> Prize Best Paper Award (CA\$207)</a><br><a href="#">[500+ submissions, Top 0.6%]</a> | <i>International level</i> |
| 2024 | <a href="#">Student Colloquium of the Montreal Centre for Interdisciplinary Research in Rehabilitation (CRIR) – People’s Choice Award (CA\$50)</a>                                                   | <i>Institutional level</i> |

### Teaching and Service

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2025 | <a href="#">Quacquarelli Symonds (QS) – Reimagine Education Bronze Award</a><br><a href="#">[Awarded as a member of the McGill SciLearn team]</a> | <i>International level</i> |
| 2025 | <a href="#">IEEE-EMBS BHI 2025 – Reviewer Recognition</a>                                                                                         | <i>International level</i> |

### Travel Awards

|      |                                                                                                                      |                            |
|------|----------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2026 | <a href="#">Catherine Berg Student Travel Award, CRIR (CA\$500)</a>                                                  | <i>Non-competitive</i>     |
| 2025 | <a href="#">Travel Award, McGill University – Centre for Research on Brain, Language and Music (CRBLM) (CA\$700)</a> | <i>Institutional level</i> |
| 2025 | <a href="#">Travel Award, Quebec Association for Physical Activity (CA\$250)</a>                                     | <i>Provincial level</i>    |
| 2024 | <a href="#">Institute Community Support Travel Award, CIHR (CA\$1,700)</a>                                           | <i>Federal level</i>       |
| 2024 | <a href="#">Travel Award, Parkinson Canada (CA\$1,000)</a>                                                           | <i>Federal level</i>       |

### Other Honors and Awards

|      |                                                                                                                                              |                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2025 | <a href="#">National Science Foundation–IEEE EMBS–Google, Young Professional NextGen Scholar Recognition (~CA\$415)</a>                      | <i>International level</i> |
| 2025 | <a href="#">McGill University 3-minute thesis competition, Guest Judge</a>                                                                   | <i>Institutional level</i> |
| 2023 | <a href="#">University of Paris – Cité M.Sc. in Informatics, “Digital Sciences” Program, Guest student member of the admission committee</a> | <i>Institutional level</i> |
| 2023 | <a href="#">University of Paris – Cité M.Sc. in Informatics, “Digital Sciences” Program, Top 3 cGPA in the Class of 2023</a>                 | <i>Institutional level</i> |
| 2023 | <a href="#">University of Paris – Cité, Summa Cum Laude [M.Sc. cGPA 4.0/4.0]</a>                                                             | <i>Institutional level</i> |
| 2021 | <a href="#">University of Paris – Cité, Magna Cum Laude [M.Sc. cGPA 3.7/4.0]</a>                                                             | <i>Institutional level</i> |

## Grant Writing

Independently developed and prepared the entire content of the following grant proposal:

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 2026 | <a href="#">Early Professionals, Inspired Careers in AgeTech (EPIC-AT) CIHR subgrant proposal “<i>What Do We Measure? Advancing the Clinical Interpretability of Wearable Mobility Biomarkers in Older Canadians with Parkinson's Disease</i>”; Mihailidis A. (PI, University of Toronto), Paquette C. (Co-PI, McGill University), Mvomo C.E. (Applicant, McGill University); [Awarded, CA\$8,000 (for translating doctoral research findings into a clinical decision support system)]</a> | <i>Federal level</i> |
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## Professional Experience

### Research Experience

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|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023–Now<br><i>Montreal, Canada</i> | <a href="#">McGill HBCL &amp; Lake Lucerne Institute DART Groups</a> , Graduate Student Researcher<br>Advisors: <a href="#">Caroline Paquette</a> and <a href="#">Chris Awai</a> |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                  | <p><i>Initiated a multi-award-winning Swiss–Canadian collaboration integrating neuroimaging, neuromotor control, AI, and wearable sensing to advance digital gait biomarkers in Parkinson’s disease.</i></p> <ul style="list-style-type: none"> <li>• Lead project conceptualization, funding acquisition, multimodal data curation and analysis, and dissemination through scientific publications and knowledge translation.</li> <li>• Mentor undergraduate and master’s students and support fellow PhD students in study conceptualization, data collection, and analysis.</li> <li>• Contribute to the broader academic community through journal and conference peer review, academic committees, and grant evaluation and writing.</li> </ul> |
| 2023<br>Lausanne,<br>Switzerland | <p><b>EPFL Rehabilitation &amp; Assistive Robotics Group</b>, Research Intern<br/> Advisor: <a href="#">Alireza Manzoori</a></p> <ul style="list-style-type: none"> <li>• Conducted multimodal analysis of inertial motion capture, exoskeleton encoder, force plate, and FSR data to identify the biomechanical factors explaining differences in assistance effectiveness.</li> <li>• Contributed to experimental design and indoor and real-world multi-terrain data collection (e.g., inertial, metabolic, heart rate, ...) to evaluate functional outcomes with and without assistance.</li> </ul>                                                                                                                                               |
| 2022-2023<br>Paris, France       | <p><b>French National Institute for Health &amp; Medical Research (UMR 1284)</b>, Research Intern<br/> Advisor: <a href="#">Kevin Lhoste</a></p> <ul style="list-style-type: none"> <li>• Prototyped a mobile health application (Flutter SDK) for fall-risk assessment in older adults, based on simple balance tests using mobile sensing and machine learning.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                          |
| 2022<br>Vitznau,<br>Switzerland  | <p><b>Lake Lucerne Institute (LLUI) DART Group</b>, Research Intern<br/> Advisor: <a href="#">Chris Awai</a></p> <p><i>Contributed to a multicenter trial integrating motion capture, virtual reality, and wearable sensing to improve balance and prevent falls in older adults.</i></p> <ul style="list-style-type: none"> <li>• Led the LLUI trial arm, including participant recruitment, multimodal data collection, preliminary analysis, and dissemination to clinical and international audiences.</li> <li>• Contributed to study conceptualization and collaboration with academic and industry partners.</li> </ul>                                                                                                                        |
| 2021<br>Compiègne, France        | <p><b>UTC Biomechanics &amp; BioEngineering (BMBI) Laboratory</b>, Research Initiation Intern<br/> Advisor: <a href="#">Khalil Ben Mansour</a></p> <ul style="list-style-type: none"> <li>• Extracurricular internship during undergraduate studies. Conducted a mini-project on the effect of walking speed on peak plantar pressure. Tasks included data acquisition, basic signal processing, interpretation of biomechanical outputs, and results dissemination to lab members.</li> </ul>                                                                                                                                                                                                                                                        |
| 2021<br>Garches, France          | <p><b>Greater Paris University Hospitals – Garches Foundation</b>, Research Initiation Intern<br/> Advisor: <a href="#">Anthony Supiot</a></p> <ul style="list-style-type: none"> <li>• Extracurricular internship during undergraduate studies. Initiated to motion capture in neurorehabilitation. Assisted researchers during quantified gait analysis sessions.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                        |

## Clinical Experience

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2021-2023<br>Évry, France | <p><b>Simone Veil Community Health Center</b>, Podiatrist Assistant<br/> Referee: Prisca Lonn (Head Podiatrist)</p> <p><i>Clinical practice (20%) in podiatric medicine conducted in parallel with the master’s degree, with a focus on sports-related conditions, aging populations, and diabetic neuropathy.</i></p> <ul style="list-style-type: none"> <li>• Activities included anamnesis, mobility assessment (e.g., gait and balance), design of custom medical devices (passive orthoses and prostheses), pharmacological and nonpharmacological treatment prescription, prevention and monitoring of complications, and communication with other healthcare professionals to support coordinated care.</li> </ul> |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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| 2020<br>Paris, France      | <b>IMSS Stade Français Paris Rugby Club</b> , Clinical Training – Sport Podiatric Medicine<br>Referee: Christophe Blanc (Head Podiatrist) <ul style="list-style-type: none"> <li>Clinical training in elite sport (private sector). Assisted a senior sport podiatrist in anamnesis, mobility assessment, design of custom medical devices (passive orthoses), nonpharmacological treatment prescription, prevention and monitoring of complications, and communication with team physicians and therapists.</li> </ul> |
| 2018-2021<br>Paris, France | <b>Greater Paris University Hospitals (AP-HP)</b> , Clinical Training – General Podiatric Medicine<br>Advisor: Yves Matillon (President of AFREP Institute at University of Paris – Cité) <ul style="list-style-type: none"> <li>Clinical training within Europe’s largest university hospital center. Hospital services included Internal Medicine, Dermatology, Rheumatology, Endocrinology, Diabetology, Rehabilitation Medicine, and Geriatrics.</li> </ul>                                                         |

### Teaching Experience

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| 2024-Now<br>Montreal, Canada | <b>McGill Office of Science Education</b> , Science Education Fellow<br>Referee: Armin Yazdani (SciLearn Program Co-Lead)<br><i>Fellow with the McGill SciLearn team, a fellowship program preparing the next generation of science instructors.</i> <ul style="list-style-type: none"> <li>Managed “SciLearn Peer-Collaboration,” a collaborative learning space supporting students in 10+ science courses. Develop and teach evidence-based neuroeducation content to 1,000+ undergraduate students and instructors in collaboration with cognitive neuroscientists and education developers. Conduct research on neuroeducation and organized student- and instructor-focused events.</li> <li>Co-founded “Étudios Ensemble,” a university-wide program supporting francophone students, and collaborated with McGill administration to support its development.</li> </ul> |
| 2024-Now<br>Montreal, Canada | <b>McGill Department of Kinesiology and Physical Education</b> , Teaching Assistant<br>Referee: Michael Creamer (Undergraduate Program Director) <ul style="list-style-type: none"> <li>Assisted the lead instructor in the “Motor Control” course (B.Sc. Kinesiology) by addressing students’ questions and difficulties with course material and assignments during office hours, mentoring students during tutorials, contributing to course and exam design, and supervising and grading written and oral examinations.</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
| 2024-2026<br>Remote          | <b>Department of Rehabilitation, AFREP, University of Paris – Cité</b> , Lecturer<br>Referee: Yves Matillon (President of AFREP Institute at University of Paris – Cité) <ul style="list-style-type: none"> <li>Developed and delivered 5×20 hours of online courses on the basics of research methodology (Bachelor of Podiatric Medicine). Activities included full course development, as well as coordination of guest lecturers.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                |

## Journal Articles

\*Equal Contribution; Mentee

### In Preparation

- [J10] **C.E. Mvomo**, T. Ogawa, C. Guedes, and C. Paquette, “Quantifying Step Length During Treadmill Walking in Parkinson’s Disease Using Inertial Sensors: A Validation Study,” 2026
- [J09] D. Leibovich, **C.E. Mvomo**, L. Higuaita, J. Wu, C. Truong, I. Nunes Loureiro, A. Potvin-Desrochers, and C. Paquette, “Freezing of Gait in Parkinson’s Disease: Does the New Definition Detect More Freezing?,” 2026 **[Preliminary results received 3<sup>rd</sup> Price Best Undergraduate Oral Presentation at Minds in Motion Conference 2026]**
- [J08] J.S.N Bedime, **C.E. Mvomo**, S. Perfetto, F. Parent-L’Ecuyer, H. Lajeunesse, I. Sierra, C. Guedes, JP Soucy, A. Potvin-Desrochers, M. Sharp, A. Rochette, A. Parent, and C. Paquette, “Brain Structural and Gait-Related Metabolic Correlates of Fatigue in Parkinson’s Disease,” 2026
- [J07] **C.E. Mvomo**, S. Neumann, J. Pohl, M. Elgendi, and C.E. Awai, “Beyond Garbage In, Garbage

Out: Can Uncertainty Become Insightful in Real-World Outcome Prediction for Parkinson's Disease?," 2026

- [J06] **C.E. Mvomo\***, J.S.N. Bedime\*, D.C. Hinton, T. Ogawa, A. Thiel, JP. Soucy, A. Potvin-Desrochers, and C. Paquette, "The Brain Network that Preserves Gait Adjustments in Parkinson's Disease," 2026

**Under Review or In Revision** (*with impact factors [IFs] at submission*)

- [J05] C. Delaney, **C.E. Mvomo**, V. Gibson, C.E. Awai, P. Chitnis, and Q. Sanders, "Assistive Technologies, Clinical Considerations, and Future Directions for Managing Freezing of Gait in Parkinson's Disease," *Clinical Parkinsonism and Related Disorders*, under review, 2026 (IF: 2.7)
- [J04] A. Parent, S. Perfetto, J.S.N. Bedime, F. Parent-L'Ecuier, I. Sierra, H. Lajeunesse, **C.E. Mvomo**, M. Sharp, A. Rochette, A. Potvin-Desrochers, and C. Paquette, "The multidimensionality of fatigue in Parkinson's disease: a qualitative study of triggers, experience and coping strategies," *Disability and Rehabilitation*, under review, 2026 (IF: 2.2)
- [J03] **C.E. Mvomo**, J.S.N. Bedime, T. Mitchell, A. Potvin-Desrochers, and C. Paquette, "A Multivariate Analysis of Frontoparietal Responses to Complex Walking in Freezing of Gait," *European Journal of Neuroscience*, in revision, 2026 (IF: 2.5)
- [J02] **C.E. Mvomo**, J.S.N. Bedime, D. Leibovich, C. Guedes, A. Potvin-Desrochers, P.C. Dixon, C.E. Awai, and C. Paquette, "[Gait-Related Digital Mobility Outcomes in Parkinson's Disease: New Insights into Convergent Validity?](#)," *IEEE Transactions on Biomedical Engineering (TBME)*, in revision, 2026 (IF: 4.5) **[invited contribution as one of the top submission at BHI 2025]**

**Published or In Press** (*with impact factors [IFs] at submission*)

- [J01] S. Neumann, **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer\*, and C.E. Awai\*, "[Visual Dependence for Postural Control is Increased in Older Adults](#)," *Aging and Disease*, 2025 (IF: 9.6)

## Conference Proceedings

All proceedings are heavily peer-reviewed and either published or in press; Mentee

- [C06] **C.E. Mvomo**, M. Elgendi, and C.E. Awai, "Epistemic Uncertainty as a Driver of Performance in Wearable Monitoring of Parkinson's Disease: A Real-World Proof-of-Concept Study," in *Converging Clinical and Engineering Research on Neurorehabilitation – Proceedings of the 7<sup>th</sup> International Conference on NeuroRehabilitation (ICNR)*, in press, 2026
- [C05] J.Y. Park, E. Morales Zaldivar, J.S.N. Bedime, **C.E. Mvomo**, F. Parent-L'Ecuier, I. Sierra, H. Lajeunesse, A. Parent, M. Sharp, A. Rochette, and C. Paquette, "Examining the Impact of Fatigue on White Matter tracts in Parkinson's Disease," in *Movement Disorders*, in press, 2026
- [C04] T. Ogawa, C. Guedes, **C.E. Mvomo**, J.S.N. Bedime, L. Hajjaji, A. Potvin-Desrochers, J. Choi, and C. Paquette, "Can people with Parkinson's disease improve gait adaptation skills via upregulation of the posterior parietal cortex?," in *Movement Disorders*, in press, 2026
- [C03] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, D. Leibovich, C. Guedes, A. Potvin-Desrochers, P.C. Dixon, C.E. Awai, and C. Paquette, "[Monitoring Parkinson's Disease In-the-Wild](#)," in *Proceedings of the IEEE-EMBS International Conference on Biomedical and Health Informatics*, 2025 **[500+ submissions, Acceptance rate <30%, 3rd Prize Best Paper Award, Top 0.6%]**
- [C02] I. Sierra, S. Perfetto, A. Potvin-Desrochers, **C.E. Mvomo**, J.S.N. Bedime, H. Lajeunesse, F. Parent-L'Ecuier, and C. Paquette. "[Influence of Fatigue in Parkinson's Disease on Variability of Gait Measures](#)," in *Movement Disorders*, 2024
- [C01] **C.E. Mvomo**, J.S.N. Bedime, I. Sierra, S. Perfetto, H. Lajeunesse, A. Potvin-Desrochers, C.E. Awai, and C. Paquette, "[Model-Based Gait Outcomes to Track Parkinson's Disease Progression?](#)," in *Movement Disorders*, 2024

## Selected Conference Posters

All peer-reviewed and limited to work not published in conference proceedings or presented orally; [Mentee](#)

- [P08] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, I. Sierra, H. Lajeunesse, F. Parent-L'Ecuyer, [T. Ogawa](#), [J.Y. Park](#), M. Sharp, M.H. Pennestri, A. Potvin-Desrochers, B.G.G. da Costa, A. Parent, and C. Paquette, "Physical Activity and Sedentary Behavior Monitoring in People with Parkinson's Disease and Fatigue," in *International Society of Posture and Gait Research (ISPGR) - World Congress*, (Maastricht, Netherlands), 2025
- [P07] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, I. Sierra, H. Lajeunesse, J.P. Soucy, A. Potvin-Desrochers, M. Sharp, A. Rochette, A. Parent, and C. Paquette, "Fatigue in Parkinson's Disease Is Linked to Enhanced Activity of the Fornix and Dysregulation of the Dorsal Attention and Central Executive Networks During Complex Walking: An [18F]-FDG PET Study," in *Annual Canadian Movement Disorders Meeting*, (Toronto, Canada), 2024
- [P06] S. Perfetto, A. Potvin-Desrochers, I. Sierra, **C.E. Mvomo**, K. Shoemaker, M. Chakravarty, and C. Paquette, "Long-term, high-level aerobic training and the age-related decline in cortical thickness," in *Society for Neurosciences (SfN) World Congress*, (Chicago, USA), 2024
- [P05] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "A Gait Model That Reflects Measures of Symptom Severity in Parkinson's Disease," in *5th Quebec Congress in Adaptation-Rehabilitation Research*, (Quebec City, Canada), 2024
- [P04] **C.E. Mvomo**, J.S.N. Bedime, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "A Gait Model That Reflects Measures of Symptom Severity in Parkinson's Disease: A Pilot Study," in *Minds in Motion Conference*, (Montreal, Canada), 2024
- [P03] J.S.N. Bedime, **C.E. Mvomo**, S. Perfetto, H. Lajeunesse, I. Sierra, A. Potvin-Desrochers, and C. Paquette, "The Influence of Cognitive Load on Locomotor Adaptation to Split-Belt Treadmill Walking in Parkinson's Disease with Freezing of Gait," in *Minds in Motion Conference*, (Montreal, Canada), 2024
- [P02] S. Neumann, **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer, and C.E. Awai, "Relative Importance of the Visual System for Balance Control Is Reflected in the Continuous Margins of Stability While Walking," in *XXIX Congress of International Society of Biomechanics (ISB)*, (Fukuoka, Japan), 2023
- [P01] **C.E. Mvomo**, D. Ravi, F. Schulte, L. Assländer, and C.E. Awai, "Does Locomotor Perturbation Training Induce a Change in Relative Sensory Contribution to Balance? A Pilot Study," in *International Neurorehabilitation Symposium (RehabWeek)*, (Rotterdam, Netherlands), 2022

## Intellectual Property

- [IP02] **C.E. Mvomo**, "[GaitHub: A digital health companion designed to democratize digital mobility biomarker analytics](#)." Software, © 2026, MIT License.
- [IP01] **C.E. Mvomo**, "[PyComplexity: A Python module for computing the Attractor Complexity Index \(ACI\), a non-linear measure of gait automaticity](#)." Software, © 2025, MIT License.

## Oral Presentations

I delivered all oral presentations listed below

### Primary Research Presentations

- 2025 | IEEE-EMBS Conference on Biomedical and Health Informatics (BHI), Atlanta, USA  
"Monitoring Parkinson's Disease In-the-Wild" **[Selected as Top 9.2% best submissions]**

### Knowledge Translation Presentations

- 2024 | TED-like Talk at NeuroLingo, Montreal, Canada **[10K+ views on YouTube]**

|      |                                                                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | “What if walking could help track the progression of Parkinson's disease?”<br>CRIR 2024 Student Colloquium, Montreal, Canada <b>[People's choice award]</b><br>“What if walking became the key to monitoring the progression of Parkinson's disease?” |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Science Education Research Presentations

|      |                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Supporting Active Learning (SALTISE) Conference, Montreal, Canada<br>“Exploring the Link Between Belief in Neuromyths and Academic Performance in Undergraduate Science Students” |
| 2025 | Skills for Success Symposium, Montreal, Canada<br>“McGill SciLearn Program” ( <i>with Kira Smith</i> )                                                                            |

### Invited Conference Presentations

|                                  |                                                                                                                                                                                                            |                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 2026<br><a href="#">[invite]</a> | AGE-WELL 11 <sup>th</sup> Annual Conference, Ottawa, Canada<br>“Wearable Clinical Decision Support Systems for Older Adults with Parkinson’s Disease: A Case Study of Decision Deferral Under Uncertainty” | <i>Federal level</i>       |
| 2026<br><a href="#">[invite]</a> | Nature Conferences “Redefining Healthcare in the Age of AI”, Paris, France<br>“When AI Defers: Uncertainty-Aware Wearable AI for Clinical Decision Support in Parkinson’s Disease”                         | <i>International level</i> |
| 2025<br><a href="#">[invite]</a> | IEEE-EMBS Conference on Biomedical and Health Informatics, Atlanta, USA<br>“NextGen Scholar Presentation: Cyrille E. Mvomo”                                                                                | <i>International level</i> |

### Teaching

#### Graduate Student Instructor (Bachelor of Podiatric Medicine)

|             |                                                                |                            |
|-------------|----------------------------------------------------------------|----------------------------|
| Winter 2026 | Introduction to Research Methods – 1 (U.E. 5.3, ~ 40 students) | University of Paris – Cité |
| Fall 2025   | Introduction to Research Methods – 2 (U.E. 5.4, ~ 20 students) | University of Paris – Cité |
| Winter 2025 | Introduction to Research Methods – 1 (U.E. 5.3, ~ 20 students) | University of Paris – Cité |
| Fall 2024   | Introduction to Research Methods – 2 (U.E. 5.4, ~ 35 students) | University of Paris – Cité |
| Winter 2024 | Introduction to Research Methods – 1 (U.E. 5.3, ~ 35 students) | University of Paris – Cité |

#### Teaching Assistant (B.Sc. Kinesiology)

|           |                                                                              |                   |
|-----------|------------------------------------------------------------------------------|-------------------|
| Fall 2026 | Motor Control (EDKP 447, ~ 80 students)                                      | McGill University |
| Fall 2025 | Motor Control (EDKP 447, ~ 80 students)                                      | McGill University |
| Fall 2024 | Motor Control (EDKP 447, ~ 70 students) ( <i>with Gleydciane Fernandes</i> ) | McGill University |

#### SciLearn Guest Lectures (in large undergraduate science courses, each from 100 to 500+ students)

Developed and delivered lectures (individually or collaboratively) on “Learning how to learn through neuroscience”

|             |                                                                                     |                   |
|-------------|-------------------------------------------------------------------------------------|-------------------|
| Fall 2026   | Introduction to Psychological Statistics (PSYC 204)                                 | McGill University |
| Fall 2026   | Introductory Organic Chemistry 1 (CHEM 212)                                         | McGill University |
| Fall 2026   | Algorithms and Data Structures (COMP 251)                                           | McGill University |
| Winter 2026 | Foundations of Programming (COMP 202) ( <i>with Jean-Christophe Boivin</i> )        | McGill University |
| Winter 2026 | Introductory Organic Chemistry 2 (CHEM 222)                                         | McGill University |
| Fall 2025   | General Chemistry 1 (CHEM 110)                                                      | McGill University |
| Winter 2025 | General Chemistry 2 (CHEM 120) ( <i>with Hilary Sweatman</i> )                      | McGill University |
| Winter 2025 | Introduction to Psychological Statistics (PSYC 204) ( <i>with Hilary Sweatman</i> ) | McGill University |

## Mentoring

Mentored 13 research projects, leading to conference and journal authorships (see publications/posters at pages 4-6).

### Graduate Students

|           |                                                                                                                                                                                                                                                        |                             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 2025-2026 | Clara Guedes, M.Sc. Student, McGill University<br><b>[2026 Canada CIHR Master Research Scholarship Recipient]</b><br>Thesis: “Split-belt Treadmill Training to Improve Gait in Parkinson’s Disease”<br>Next: Medical Student at Université de Montréal | Biomechanics & Neuroscience |
| 2024-2026 | Jae Young Park, M.Sc. Student, McGill University<br>Thesis: “Fatigue-Related White Matter Changes in Parkinson’s Disease”<br>Next: Medical Student at University of Aberdeen                                                                           | Biomechanics & Neuroscience |
| 2024-2026 | Tomoha Ogawa, M.Sc. Student, McGill University<br>Thesis: “Parietal Cortex Neuromodulation to Improve Gait in Parkinson”                                                                                                                               | Biomechanics & Neuroscience |
| 2026      | Florian Gourdin, Visiting M.Eng. Student, Polytech Marseille<br>Project: “Parietal Cortex Upregulation to Improve Freezing of Gait”<br>Next: Final Year M.Eng. Student at Polytech Marseille                                                           | Biomedical Engineering      |

### Undergraduate Students

|           |                                                                                                                                                                                                                                                                                      |                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 2026      | Daniela Garcia Garcia, Visiting B.Sc. Student, Universidad de Monterrey<br><b>[2026 Mitacs Globalink Summer Research Internship Program]</b><br>Project: “Inertial Step Length Validation on a Treadmill in Parkinson”<br>Next: Final Year B.Sc. Student at Universidad de Monterrey | Biomedical Engineering |
| 2026      | Marilena Sahlas-Roy, B.Sc. Student, McGill University<br>Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease”<br>Next: 2 <sup>nd</sup> Year B.Sc. Student, McGill University                                                                                         | Kinesiology            |
| 2026      | Yasmine Abouelfareh, B.Sc. Student, McGill University<br>Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease”<br>Next: 2 <sup>nd</sup> Year B.Sc. Student, McGill University                                                                                         | Kinesiology            |
| 2026      | Tatiana Amaya Quintero, B.Sc. Student, McGill University<br>Project: “Annotating Ziegler Test Recordings in Parkinson’s Disease”<br>Next: Final Year B.Sc. Student, McGill University                                                                                                | Kinesiology            |
| 2026      | Justin Wu, B.Sc. Student, McGill University<br>Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease”<br>Next: 3 <sup>rd</sup> Year B.Sc. Student, McGill University                                                                                               | Kinesiology            |
| 2026      | Ines Nunes Loureiro, B.Sc. Student, McGill University<br>Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease”<br>Next: Final Year B.Sc. Student, McGill University                                                                                               | Kinesiology            |
| 2026      | Ciara Truong, B.Sc. Student, McGill University<br>Project: “Annotating Freezing of Gait Recordings in Parkinson’s Disease”<br>Next: Senior Leader at Vancouver Champlain Heights Community Centre                                                                                    | Kinesiology            |
| 2024-2026 | Dahlia Leibovich, B.Sc. Student (Honors), McGill University<br><b>[2026 Canada CIHR Master Research Scholarship Recipient]</b><br>Thesis: “Evaluating Freezing of Gait Severity in Parkinson’s Disease”<br>Next: M.Sc. Student in Kinesiology at the University of Toronto           | Kinesiology            |
| 2024      | Laura Higueta, B.Sc. Student, Universidad Autónoma de Occidente<br><b>[2024 Mitacs Globalink Summer Research Internship Program]</b><br>Thesis: “Detecting Freezing of Gait in Parkinson’s Disease”<br>Next: Final Year B.Sc. Student at Universidad Autónoma de Occidente           | Biomedical Engineering |

## Media Coverage

|      |                                                                                                                                                  |                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2026 | <a href="#">Funded Research – 2026 Research competition</a>                                                                                      | Parkinson Canada |
| 2026 | <a href="#">SciLearn: Learning how to learn</a>                                                                                                  | The Tribune      |
| 2026 | <a href="#">KPE PhD Candidate Cyrille Mvomo Honored with IEEE-EMBS BHI 2025 Best Paper Award and NSF-EMBS-Google NextGen Scholar Recognition</a> | McGill News      |
| 2024 | <a href="#">Results – CRIR 2024 Student Colloquium</a>                                                                                           | CRIR News        |

## Entrepreneurship

|          |                                                                                                                                                                                                                                                                                                                         |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-Now | Co-founder and manager of “Études Ensemble,” McGill’s first university-wide program providing undergraduate students with a space to study in French and build connections. The program attracted students from across the university and secured support from McGill’s administration and Office of Science Education. |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Professional Service

### Undergraduate Theses Defense Committee Member

|      |                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Xavier Surget, AFREP Institute, University of Paris – Cité<br>Thesis: “Effect of Foot Orthoses on Pressure Points of a Flexible Flat Foot” |
| 2026 | Flavie Tran Dinh, AFREP Institute, University of Paris – Cité<br>Thesis: “Podiatric Management of Blisters in Desert Environments”         |
| 2026 | Coline Vollant, AFREP Institute, University of Paris – Cité<br>Thesis: “Effect of Running on Foot Posture in Novice and Trained Runners”   |
| 2025 | Zoé Imbault, AFREP Institute, University of Paris – Cité<br>Thesis: “The Impact of Raynaud’s Phenomenon on Skiers”                         |
| 2025 | Rachelle Olivette, AFREP Institute, University of Paris – Cité<br>Thesis: “Impact of Foot Orthoses on Low Back Pain”                       |
| 2025 | Hugo Forgez, AFREP Institute, University of Paris – Cité<br>Thesis: “Running Shoes : Prevention and Treatment of Plantar Fasciitis”        |

### Conference Organizer

|      |                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | McGill Undergraduate Science Showcase, Member of the Organizing Team<br><i>Conference aimed at providing undergraduate science students with the opportunity to present their research.</i> |
| 2025 | McGill Undergraduate Science Showcase, Member of the Organizing Team<br><i>Conference aimed at providing undergraduate science students with the opportunity to present their research.</i> |

### Workshop Organizer

|        |                                                                                                                               |
|--------|-------------------------------------------------------------------------------------------------------------------------------|
| Fall   | SciLearn Workshop at Discover McGill, McGill University, Facilitator (with J.C. Boivin)                                       |
| 2026   | <i>Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation.</i>                     |
| Winter | SciLearn Workshop “Prep for Finals,” McGill University, Facilitator (with J.C. Boivin)                                        |
| 2026   | <i>Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.</i>      |
| Winter | SciLearn Workshop “MidTerm Bounce-Back,” McGill University, Facilitator (with J.C. Boivin)                                    |
| 2026   | <i>Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.</i> |
| Fall   | SciLearn Orientation Week Workshop, McGill University, Facilitator                                                            |
| 2025   | <i>Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation.</i>                     |
| Fall   | SciLearn Training Workshop “How Students Learn,” McGill University, Facilitator (with A. Yazdani)                             |
| 2025   | <i>Workshop aimed at providing teaching assistants in Chemistry with basic knowledge of neuroeducation.</i>                   |
| Winter | SciLearn Workshop “Prep for Finals,” McGill University, Facilitator (with H. Sweatman)                                        |

- 2025 | *Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.*
- Winter 2025 | SciLearn Workshop “MidTerm Bounce-Back,” McGill University, Facilitator (with H. Sweatman)
- 2025 | *Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.*
- Fall 2024 | SciLearn Workshop “Prep for Finals,” McGill University, Facilitator (with H. Sweatman)
- 2024 | *Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation for the finals.*
- Fall 2024 | SciLearn Workshop “MidTerm Bounce-Back,” McGill University, Facilitator (with H. Sweatman)
- 2024 | *Workshop aimed at providing undergraduate science students with basic knowledge of neuroeducation after the mid-terms.*

### Conference Reviewer

- 2026 | IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'26)  
*Review of 5 articles on Precision Health and AI in Parkinson.*
- 2025 | IEEE-EMBS International Conference on Biomedical and Health Informatics (BHI'25)  
*Review of 6 articles on Precision Health and AI. [Reviewer Recognition]*

### Journal Reviewer

- 2024 | Medicine & Science in Sports & Exercise  
*Ad hoc review (via supervisor) of an article on wearable technologies for monitoring mobility disorders in Parkinson.*

### Academic Committee Member

- 2024-Now | Committee on Innovative Practices in Research Ethics in Rehabilitation (PIER), CRIR  
*Student member of a committee that supports CRIR research teams in addressing emerging research ethics challenges, such as participant recruitment, digital data management, and support tools.*
- 2023-Now | Research Committee, Rehabilitation Department, University of Paris – Cité  
*Alternate member of a committee dedicated to promoting research within the newly created department.*

### Social Mobilization

- 2024 | Patient Engagement Workshop, Parkinson Canada  
*Participation in a workshop aimed at improving interaction between researchers, patients, clinicians and industrial partners to enhance patient engagement in Parkinson's research in Canada.*
- 2022 | Hackathon, HackaHealth Zürich  
*Participated in hackathon to develop solutions for people with disabilities.*
- 2017 | Fundraising Events, Or Bleu  
*Participation in soccer tournaments to finance the construction of water wells in Mali.*

## Professional Development

- 2026-Now | EPIC-AT Training  
*Interdisciplinary training in AgeTech innovation to translate research findings into digital solutions for older Canadians.*
- 2024-Now | McGill Science Education Fellow Program  
*Training to become next-generation science faculty through lectures/workshops on science education and faculty mentoring.*
- 2024 | V1 Studio Researchers and Entrepreneurs Network Program (RCE)  
*Entrepreneurship training for researchers to translate research findings into real-world solutions.*
- 2023 | CIHR Integrating Sex and Gender into Health Research  
*Training in integrating sex and gender considerations into health research design, methods, analysis, and interpretation.*
- 2023 | Government of Canada TCPS 2 CORE-2022 (Research Ethics)  
*Training in the ethical conduct of research involving humans under Canada's Tri-Council Policy Statement (TCPS 2).*
- 2022 | Motek Medical Gait Analysis & Rehabilitation using CAREN (Operator Level 1)  
*Training on fully immersive multimodal motion analysis (e.g., optical, inertial, force plates, EMG, ...) using CAREN.*
- 2021 | French Ministry of Health AFGSU 1 and 2 (emergency care)  
*Training in first aid for healthcare professionals, including recognition and initial management of medical emergencies.*

## Professional Affiliations & Licensure

|           |                                                                                         |
|-----------|-----------------------------------------------------------------------------------------|
| 2025-Now  | Student Member, Institute of Electrical and Electronics Engineers (IEEE)                |
| 2025-Now  | Student Member, IEEE Engineering in Medicine and Biology Society (EMBS)                 |
| 2024-Now  | Student Member, Movement Disorders Society (MDS)                                        |
| 2023-Now  | Student Member, Montreal Centre for Interdisciplinary Research in Rehabilitation (CRIR) |
| 2023-Now  | Student Member, CRBLM, Faculty of Medicine and Health Sciences, McGill University       |
| 2023-2026 | Student Member, Quebec BioImaging Network (QBIN)                                        |
| 2021-2023 | Member, French Podiatric Medicine Society (SOFPOD)                                      |
| 2021-2023 | Registered Podiatrist, French National Council of Podiatrists (ONPP)                    |

## References

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